

Data Communication and Computer Networks

Project Title:

Blockchain for Networking: A Basic Simulation for Secure Transactions in IoT/Mobile Networks

Student's Details:

Student	Roll no	Email
Rimsha Hassan	SU91-BIETM-F23-037	su91-bietm-f23-037@superior.edu.pk

Teacher Details:

Supervised by:	Ms. Eisha Nawaz
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Project Overview:

This project represents a complete blockchain-based network architecture designed in Cisco Packet Tracer.

The topology consists of **three interconnected segments**:

- Left Side LAN
- IoT Zone
- Right Side LAN.

All three segments communicate through routers that are linked together, allowing smooth data flow across the entire system and enabling connectivity with an external WAN/Cloud for global blockchain synchronization.

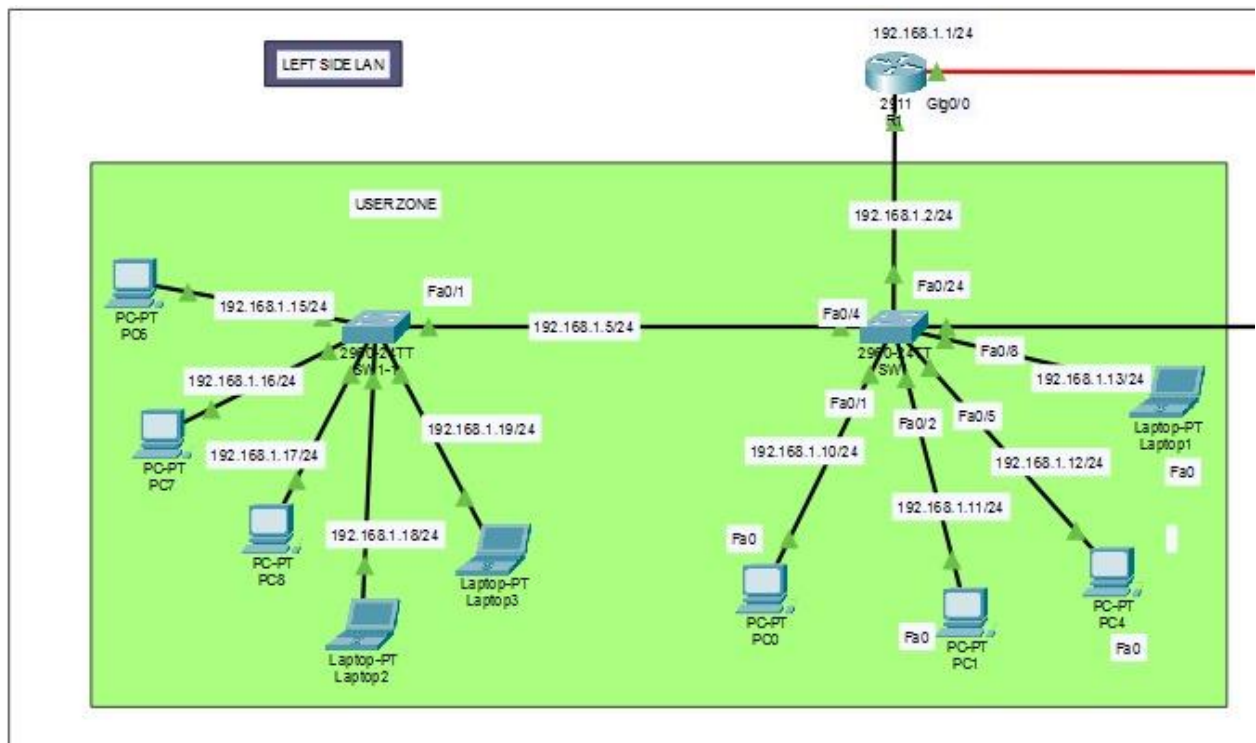
Left Side LAN: (Blockchain Node Network)

The left segment of the topology contains multiple PCs and laptops connected through access switches and a distribution switch.

These devices act as **blockchain nodes**, responsible for:

- Storing and maintaining the distributed ledger
- Validating and exchanging blockchain data
- Participating in peer-to-peer communication
- Synchronizing blocks with other LANs through the router

This entire left-side LAN forms the main **blockchain full-node area**, supporting decentralization and reliable ledger availability.



IoT Zone: (Smart Device Environment)

The central part of the network is the **IoT zone**, where various smart devices are connected through an IoT switch.

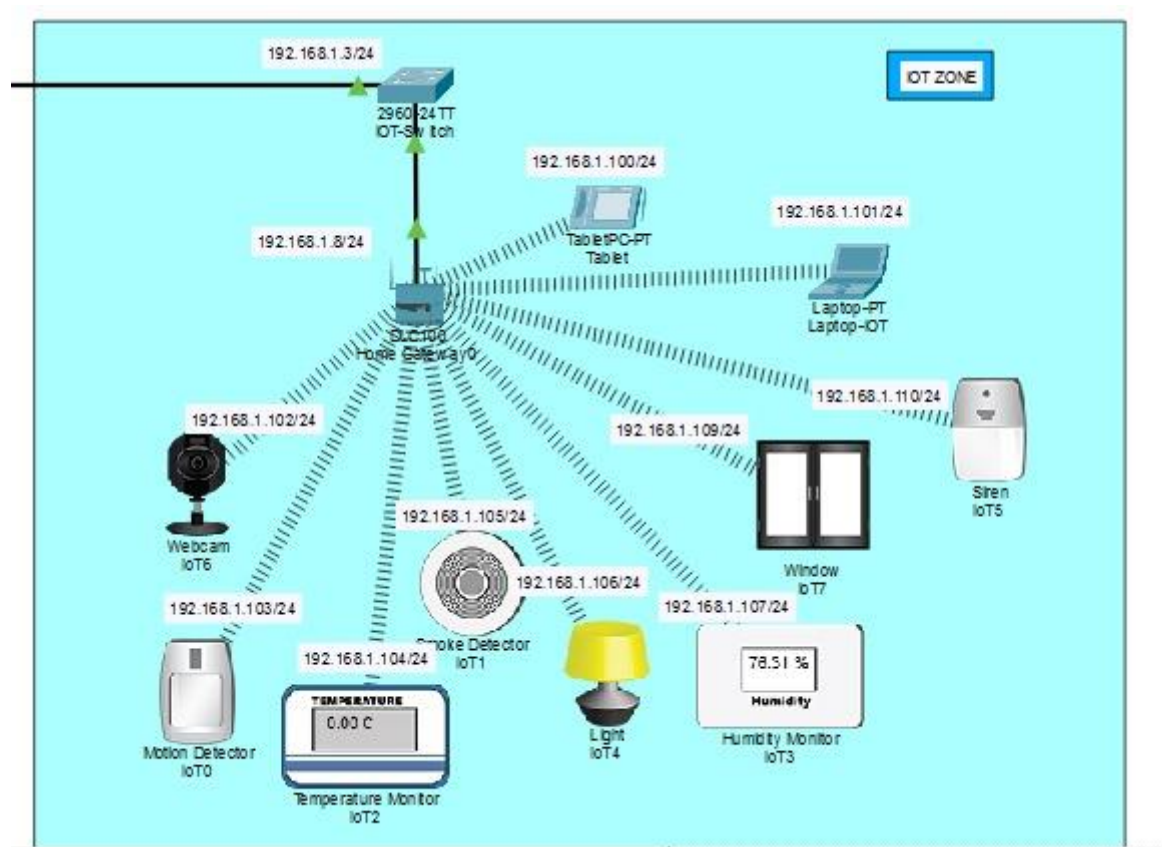
This segment includes:

- Temperature and humidity sensors
- Motion and smoke detectors
- Smart window
- Smart light and siren
- Webcam
- IoT tablet and laptop

These devices generate real-time data and alerts.

Their traffic is routed through the OT router, allowing the IoT environment to communicate with both the Left LAN (blockchain nodes) and the Right LAN.

This demonstrates **integration of IoT data into a secure blockchain-backed environment**.



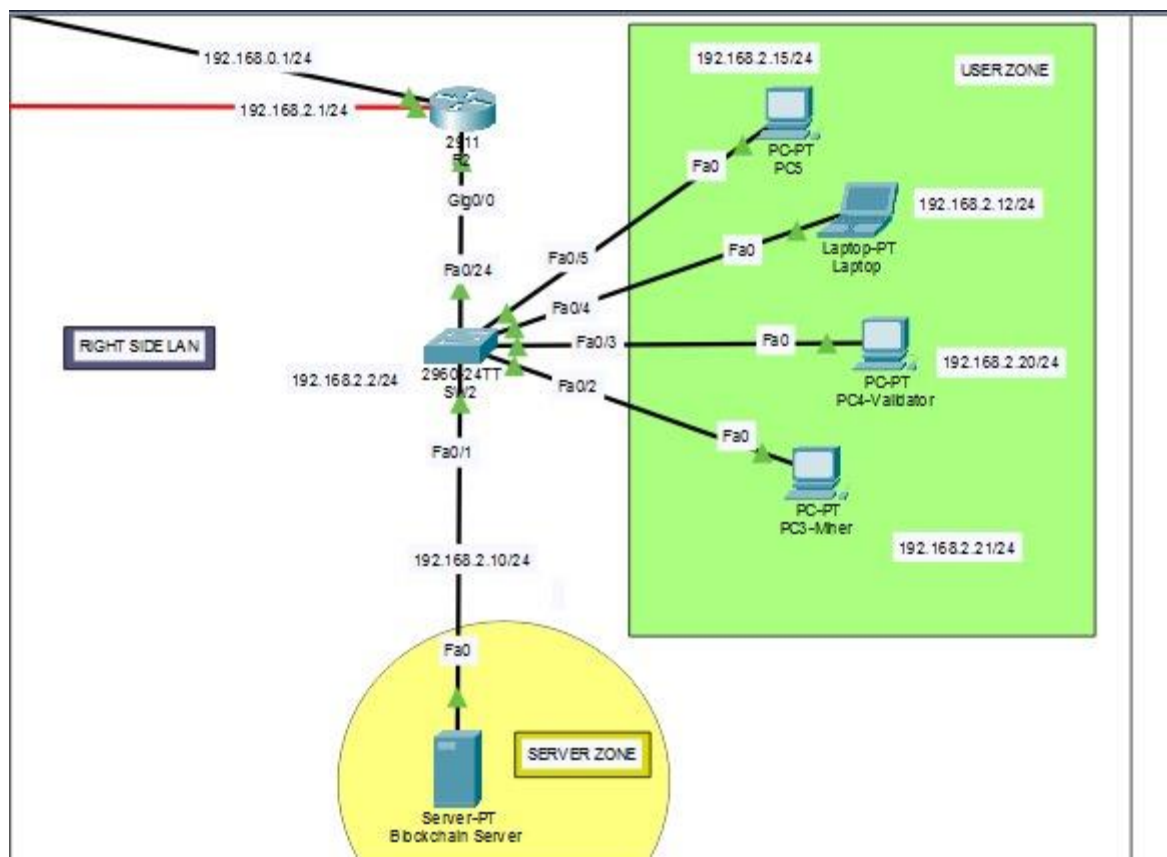
Right Side LAN: (User & Processing Network)

The right segment consists of PCs connected to a local switch and router.

These devices function as:

- General user systems
- Additional blockchain participants
- Workstations communicating with IoT devices and the left-side blockchain nodes

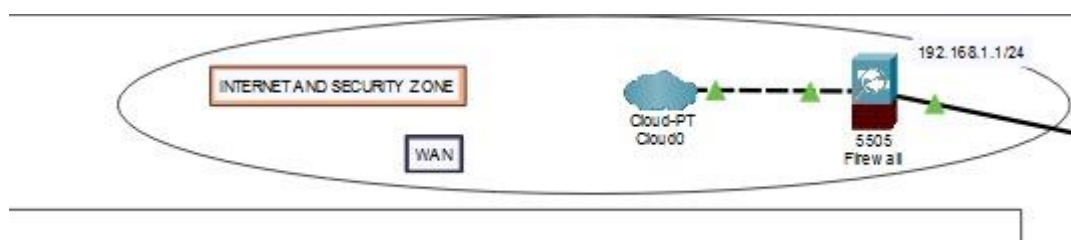
This LAN handles blockchain communication, data processing, and inter-LAN connectivity. It forms the third major section of the network, supporting system-wide communication and distributed operations.



WAN / Cloud Connectivity:

All routers from the Left LAN, IoT Zone, and Right LAN are connected to a WAN cloud, enabling:

- External blockchain synchronization.
- Internet access for all devices.
- Communication with global nodes.
- Realistic distributed blockchain behavior.



Integrated System Functioning:

All segments **Left LAN**, **IoT Zone**, and **Right LAN** are fully interconnected through routers. Devices can seamlessly exchange data, allowing:

- IoT sensors → OT Router → Left/Right LAN
- Blockchain nodes in both LANs to sync and validate data
- Entire topology to access global blockchain via the WAN

This results in a complete, secure, and fully functional blockchain-IoT integrated network.

